Daniel Kushnir kushnir.daniel05@gmail.com
Hao Wang (corresponding author) hoguewang@gmail.com
Rutgers University

October 14, 2026

Editors-in-Chief
Journal of Machine Learning Research

Dear Editors,

We submit our manuscript *TSFM-Lens: Controlled, Causal and Confirmatory Interpretability for Time-Series Foundation Models* for consideration as a regular paper in the Journal of Machine Learning Research.

Time-series foundation models are now used zero-shot for forecasting, but the interpretability methods developed for language models do not carry over directly. Tokens span different lengths of time in different architectures, real series carry no labels that could validate a probe, and the causal effects of internal features are small enough that uncontrolled measurements mislead. The manuscript presents an open-source system built around these three problems and the results of applying it to seven models.

The main contributions are:

1. **A comparative interpretability system.** A measured token-to-time alignment puts architectures with tokens of 1 to 96 time steps on a common axis, and a leakage-audited synthetic benchmark with exact generative ground truth and sealed development and private splits supplies the data. A 19-stage pipeline produces one report in which every claim carries its evidence class.
2. **A causal concept methodology.** A sparse-autoencoder feature counts as causal only if removing it changes the forecast more than removing the same per-token amount along a matched random direction on the same series, after reach controls confirm that the intervention lands. Concepts are then compared across models in two separate senses: whether models respond to the same inputs, and whether the effects on the forecast agree.
3. **Known-answer validation.** Each null and test is first run on a forecaster with planted concepts and decoys, against a pass criterion fixed in advance. The profile-matched null passes (false-positive rate at most 0.003, sensitivity 0.70–1.00); a uniform null, a more powerful per-feature test and a subspace redesign fail and are reported as such.
4. **A seven-model study confirmed once on two corpora** (TimesFM, Chronos-2, Sundial, Chronos-Bolt, Chronos-T5, Timer and Time-MoE). Claims were registered on development data and tested once on sealed private data: 43 of 82 replicable claims confirm on the first corpus and 41 of 63 on a second, three times larger corpus. Different architectures select the same input series for a concept: all 20 registered transfer claims confirm on both private splits and on 800 raw real-world windows. Their causal machinery, however, is organized by model, and no concept spans more than three of the seven models. Activation prominence is a poor guide to causal importance: 83.5% of prominent features have no held-out effect. Single causal features survive mean ablation (69 of 91) but exceed random directions drawn

from the activation distribution only 1.29-fold, and neither more data, a more powerful test nor gradient-selected candidates changes this. Depth-resolved reaction profiles replicate for most corruptions, with one that changes sign across corpora.

We report negative results with the same weight as positive ones, since several of the instruments we first used would have produced confident false positives.

We believe the paper suits JMLR because its contribution is methodological as much as empirical: it shows that interpretability claims about forecasting models need known answers, matched nulls and held-out confirmation, and it supplies a tool that makes these the default. All code, the benchmark generators, a complete seven-model example run with its held-out confirmation, and a reproduction command for every reported number are public under the MIT license at `https://github.com/DanK10097110/TSFM-Interp`, archived at doi:10.5281/zenodo.23267462 (version 1.1.0) and installable from PyPI.

The manuscript is not under review elsewhere. A preprint is available on arXiv at [arXiv identifier]. We plan to describe the software itself separately in a short paper for the JMLR Machine Learning Open Source Software track; that paper covers the toolkit's design and availability, not the scientific results reported here.

Use of AI tools: Claude (Anthropic) wrote most of the repository's code and tests, ran the experiments and helped draft and edit the manuscript, under the authors' direction. The research questions, ideas and general design choices are the authors'. The authors reviewed the implementation details Claude proposed, checked every reported result and take full responsibility for the content.

Suggested action editors: [names]. Suggested reviewers: [names, affiliations, emails]. We have no conflicts of interest to declare.

Sincerely,

Daniel Kushnir and Hao Wang
Rutgers University